<u>AI ASSISTED CODING</u>

ASSIGNMENT - 9.1

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BATCH - 28

Problem 1:
Consider the following Python function: def find_max(numbers): return max(numbers)

Task:

• Write documentation for the function in all three formats:

(a) Docstring
(b) Inline comments
(c) Google-style documentation

• Critically compare the three approaches. Discuss the advantages, disadvantages, and suitable use cases of each style.

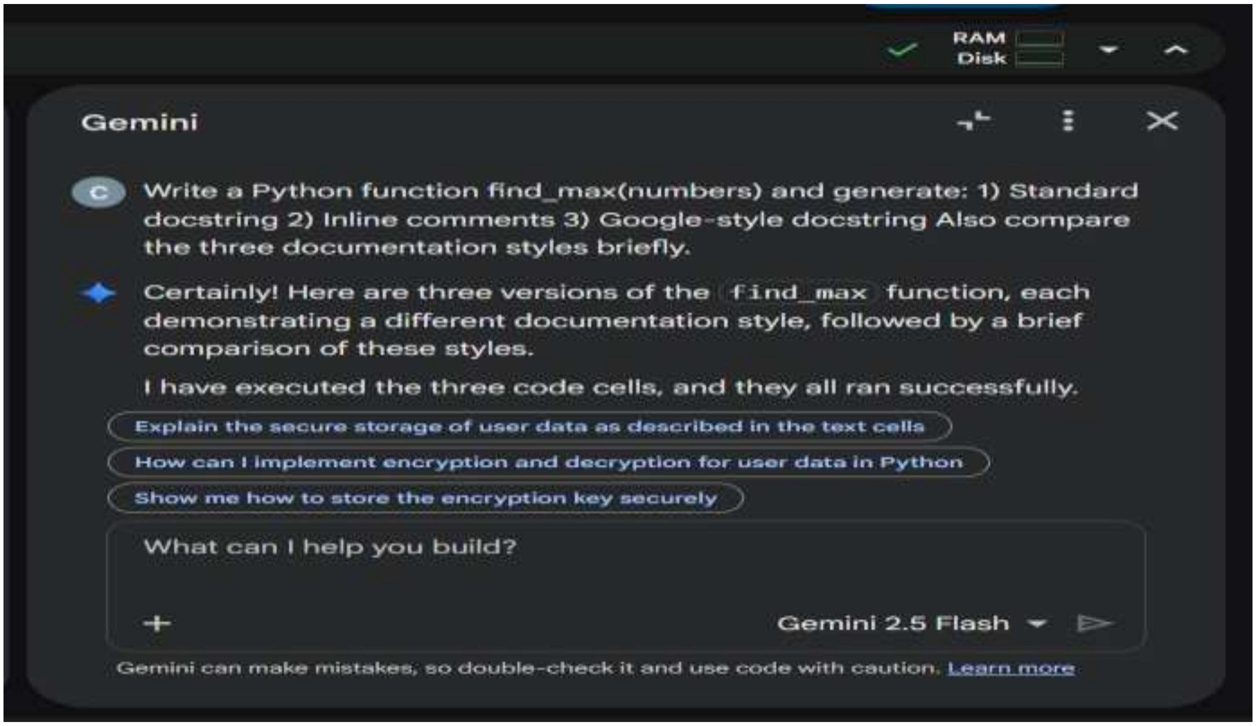
• Recommend which documentation style is most effective for a mathematical utilities library and justify your answer.

Initial code:

```
def find_max(numbers):
  return max(numbers)

print(find_max([1, 3, 7, 9, 2]))
```

Prompt:
Write a Python function find_max(numbers) and generate: 1) Standard docstring 2) Inline comments 3) Google-style docstring Also compare the three documentation styles briefly.



Code:

```
def find_max_standard_docstring(numbers):
    """
    Finds the maximum number in a list of numbers.

    Args:
        numbers (list): A list of numerical values.

    Returns:
        int or float: The maximum value in the list.
    """     if not
numbers:
        raise ValueError("Input list cannot be empty")
    return max(numbers)

print("Standard Docstring Example:", find_max_standard_docstring([1, 5, 2,
9, 3]))
```

```python
def find_max_inline_comments(numbers):
    # Checks if the input list is empty
    if not numbers:
        # Raises an error if the list is empty
        raise ValueError("Input list cannot be empty")      # Returns
the maximum value from the list using the built-in max()
function
    return max(numbers)

print("Inline Comments Example:", find_max_inline_comments([10, 4, 7, 12,
1]))
```

```python
def find_max_google_docstring(numbers):
    """     Finds the maximum number in a list of
numbers.

    Args:
        numbers (list): A list of numerical values.

    Returns:
        int or float: The maximum value in the list.

    Raises:
        ValueError: If the input list is empty.

    Example:
        >>> find_max_google_docstring([1, 5, 2, 9, 3])
        9
        >>> find_max_google_docstring([])
        ValueError: Input list cannot be empty
    """
    if not numbers:
        raise ValueError("Input list cannot be empty")
    return max(numbers)

print("Google-style Docstring Example:", find_max_google_docstring([20,
```

```
15, 25, 18]))
```

Output:

```
Standard Docstring Example: 9
Inline Comments Example: 12
Google-style Docstring Example: 25
```

Recommendation — Mathematical Utilities Library

Use Google-style docstrings

Justification:

Mathematical libraries are reused widely and often auto-documented. Google-style docstrings provide structured parameters and return descriptions that integrate well with documentation tools and improve readability.

Problem 2: Consider the following Python function: def login(user, password, credentials): return credentials.get(user) == password Task:

1. Write documentation in all three formats.
2. Critically compare the approaches.
3. Recommend which style would be most helpful for new developers onboarding a project, and justify your choice.

Initial code:

```
def login(user, password, credentials):
  return credentials.get(user) == password
```

Prompt:

Document a Python login() function using docstring, inline comments, and Google-style format. Compare which helps new developers most.

Code:

```
def login_standard_docstring(user, password, credentials):
"""

    Authenticates a user by checking their provided password against

stored credentials.

Args:

        user (str): The username attempting to log in.

        password (str): The password provided by the user.

        credentials (dict): A dictionary where keys are usernames (str)

and values are their passwords (str).


Returns:

        bool: True if the user's password matches the stored credentials,

False otherwise.

"""
```

```
    return credentials.get(user) == password


# Example usage:

user_credentials = {'admin': 'password123', 'guest': 'guestpass'}

print("Standard Docstring Login (admin, correct pass):",

login_standard_docstring('admin', 'password123', user_credentials))

print("Standard Docstring Login (admin, wrong pass):",

login_standard_docstring('admin', 'wrongpass', user_credentials))


def login_inline_comments(user, password, credentials):

    # Retrieve the stored password for the given user from the credentials

dictionary
.

    # If the user does not exist, credentials.get(user) will return None.

    stored_password = credentials.get(user)


    # Compare the provided password with the stored password.

    # If both are valid and match, authentication is successful.

    return stored_password == password
```

```
# Example usage:

user_credentials = {'admin': 'password123', 'guest': 'guestpass'}

print("Inline Comments Login (guest, correct pass):",

login_inline_comments('guest', 'guestpass', user_credentials))

print("Inline Comments Login (guest, wrong pass):",

login_inline_comments('guest', 'badpass', user_credentials))
```

```
def login_google_docstring(user: str, password: str, credentials: dict) ->

bool:

    """

    Authenticates a user against a dictionary of stored credentials.


    This function checks if the provided username exists in the

`credentials` dictionary

    and if the `password` matches the stored password for that user.


    Args:
```

```
        user: The username attempting to log in.
        password: The password provided by the user.
        credentials: A dictionary mapping usernames (str) to their
passwords (str).

Returns:
        True if the username exists and the password matches, False
otherwise.

Example:
        >>> user_creds = {'testuser': 'secretpass'}
        >>> login_google_docstring('testuser', 'secretpass', user_creds)
True
        >>> login_google_docstring('testuser', 'wrongpass', user_creds)
False
        >>> login_google_docstring('nonexistent', 'anypass', user_creds)
```

```python
False

    """

    return credentials.get(user) == password


# Example usage:

user_credentials = {'manager': 'securepwd', 'developer': 'devpass'}

print("Google-style Docstring Login (manager, correct pass):",

login_google_docstring('manager', 'securepwd', user_credentials))

print("Google-style Docstring Login (developer, wrong pass):",

login_google_docstring('developer', 'wrongpass', user_credentials))
```

Output:

```
Standard Docstring Login (admin, correct pass): True
Standard Docstring Login (admin, wrong pass): False
Inline Comments Login (guest, correct pass): True
Inline Comments Login (guest, wrong pass): False
Google-style Docstring Login (manager, correct pass): True
Google-style Docstring Login (developer, wrong pass): False
```

Recommendation — For New Developers

Best: Google-style docstrings

Justification to write:

New developers benefit from structured parameters and return descriptions. Google-style docstrings make onboarding easier because expectations and data types are clearly defined.

Problem 3: Calculator (Automatic Documentation Generation) Task: Design a Python module named calculator.py and demonstrate automatic documentation generation.

Instructions:

1. Create a Python module calculator.py that includes the following functions, each written with appropriate docstrings:
o add(a, b) – returns the sum of two numbers o subtract(a, b) – returns the difference of two numbers o multiply(a, b) – returns the product of two numbers o divide(a, b) – returns the quotient of two numbers 2. Display the module documentation in the terminal using Python's documentation tools.

3. Generate and export the module documentation in HTML format using the pydoc utility, and open the generated HTML file in a web browser to verify the output

Prompt:

Create calculator.py module with add, subtract, multiply, divide functions using proper docstrings compatible with pydoc.

Code:

```
"""
Calculator module providing basic arithmetic operations.


This module contains functions for performing addition, subtraction,
```

```
multiplication, and division operations.
"""


def add(a, b):
    """
    Add two numbers.

    Args:
        a (float): The first number.
        b (float): The second number.

    Returns:
        float: The sum of a and b.
    """
    return a + b


def subtract(a, b):
```

```
"""

    Subtract two numbers.

Args:

        a (float): The first number.

        b (float): The second number.

    Returns:

        float: The difference of a and b.

"""
```

```
    return a - b

def multiply(a, b):

"""

    Multiply two numbers.
```

```
Args:

        a (float): The first number.

        b (float): The second number.


    Returns:

        float: The product of a and b.

"""

    return a * b



def divide(a, b):

"""

    Divide two numbers.


Args:

        a (float): The dividend.

        b (float): The divisor (must not be zero).
```

```
    Returns:
        float: The quotient of a divided by b.

Raises:
        ValueError: If b is zero.
"""
    if b == 0:
        raise ValueError("Cannot divide by zero")
    return a / b

# Example usage:
if __name__ == "__main__":
    num1 = 10
    num2 = 5

    print(f"Addition: {num1} + {num2} = {add(num1, num2)}")
    print(f"Subtraction: {num1} - {num2} = {subtract(num1, num2)}")
    print(f"Multiplication: {num1} * {num2} = {multiply(num1, num2)}")
```

```
    print(f"Division: {num1} / {num2} = {divide(num1, num2)}")
```

Output:

```
Addition: 10 + 5 = 15
Subtraction: 10 - 5 = 5
Multiplication: 10 * 5 = 50 Division:
10 / 5 = 2.0
```

Command:

python -m pydoc calculator

Output:

```
PS C:\Users\rohit\OneDrive\Documents\SRU\ai_code\lab_9.1> python
-m pydoc calculator
>>
Help on module calculator:

NAME
calculator

FUNCTIONS
add(a, b)
        Add two numbers.

        Args:
            a: First number
b: Second number

        Returns:
            The sum of a and b

    divide(a, b)
        Divide two numbers.
```

```
        Args:
            a: First number (dividend)
b: Second number (divisor)

        Returns:
            The quotient of a divided by b

        Raises:
            ZeroDivisionError: If b is zero

    multiply(a, b)
        Multiply two numbers.

        Args:
            a: First number
b: Second number

        Returns:
            The product of a and b

    subtract(a, b)
        Subtract two numbers.

        Args:
            a: First number
b: Second number

        Returns:
            The difference of a and b

FILE
c:\users\rohit\onedrive\documents\sru\ai_code\lab_9.1\calculator.p
y
```

Command for html export: python -m pydoc -w calculator

```
PS C:\Users\rohit\OneDrive\Documents\SRU\ai_code\lab_9.1> python
-m pydoc -w calculator
>>  wrote
calculator.html
```

Output (calculator.html):

```
                 index
calculator c:\users\rohit\onedrive\documents\sru\ai_code\lab_9.1\calculator.py

Functions

    add(a, b)
        Add two numbers.

        Args:
          a: First number
          b: Second number

        Returns:
          The sum of a and b

    divide(a, b)
        Divide two numbers.

        Args:
          a: First number (dividend)
          b: Second number (divisor)

        Returns:
          The quotient of a divided by b

        Raises:
          ZeroDivisionError: If b is zero

    multiply(a, b)
        Multiply two numbers.

        Args:
          a: First number
          b: Second number

        Returns:
          The product of a and b

    subtract(a, b)
        Subtract two numbers.

        Args:
          a: First number
          b: Second number

        Returns:
          The difference of a and b
```

Problem 4: Conversion Utilities Module Task:

1. **Write a module named conversion.py with functions:**
o decimal_to_binary(n) o
binary_to_decimal(b) o
decimal_to_hexadecimal(n)
2. **Use Copilot for auto-generating docstrings.**
3. **Generate documentation in the terminal.**
4. **Export the documentation in HTML format and open it in a browser**

Prompt:
Generate conversion.py module with decimal_to_binary, binary_to_decimal, decimal_to_hexadecimal functions and detailed docstrings.

Code:
```
"""

Conversion module for number system conversions.



This module provides functions to convert between different number systems

including decimal, binary, and hexadecimal representations.

"""



def decimal_to_binary(decimal_num):

    """

    Convert a decimal number to its binary representation.



    Args:
```

```
        decimal_num (int): A non-negative integer to convert to binary.

    Returns:

        str: The binary representation of the decimal number (without '0b'
prefix).

    Raises:

        ValueError: If the input is negative.
```

```
    Example:

        >>> decimal_to_binary(10)

'1010'

        >>> decimal_to_binary(255)

'11111111'

"""

    if decimal_num < 0:
```

```python
        raise ValueError("Input must be a non-negative integer")

    return bin(decimal_num)[2:]



def binary_to_decimal(binary_str):

"""

    Convert a binary string to its decimal representation.


Args:

        binary_str (str): A binary number as a string (e.g., '1010').


    Returns:

        int: The decimal equivalent of the binary number.


Raises:

        ValueError: If the input contains non-binary characters.


    Example:

        >>> binary_to_decimal('1010')
```

```
        10
        >>> binary_to_decimal('11111111')
255
"""
    try:
        return int(binary_str, 2)
    except ValueError:
        raise ValueError("Input must contain only binary digits (0 and 1)")


def decimal_to_hexadecimal(decimal_num):
"""
    Convert a decimal number to its hexadecimal representation.
Args:
        decimal_num (int): A non-negative integer to convert to hexadecimal.
```

```
    Returns:

        str: The hexadecimal representation of the decimal number (without '0x'
prefix).

    Raises:

        ValueError: If the input is negative.

    Example:

        >>> decimal_to_hexadecimal(255)

        'ff'

        >>> decimal_to_hexadecimal(4095)

        'fff'

    """

    if decimal_num < 0:

        raise ValueError("Input must be a non-negative integer")

    return hex(decimal_num)[2:]

# Example usage:
```

```python
if __name__ == "__main__":

    print(decimal_to_binary(10))  # Output: '1010'

    print(binary_to_decimal('1010'))  # Output: 10

    print(decimal_to_hexadecimal(255))  # Output: 'ff'
```

Output:

```
1010
10 ff
```

Command:

python -m pydoc conversion

Output:

```
PS C:\Users\rohit\OneDrive\Documents\SRU\ai_code\lab_9.1> python
-m pydoc conversion
Help on module conversion:

NAME    conversion - Conversion module for number system
conversions.

DESCRIPTION
    This module provides functions to convert between
different number systems    including decimal, binary, and
hexadecimal representations.

FUNCTIONS
    binary_to_decimal(binary_str)
        Convert a binary string to its decimal representation.

        Args:
            binary_str (str): A binary number as a string (e.g.,
'1010').

        Returns:
            int: The decimal equivalent of the binary number.

        Raises:
            ValueError:  If  the  input  contains  non-binary
characters.

        Example:
            >>> binary_to_decimal('1010')
            10
            >>> binary_to_decimal('11111111')
            255

    decimal_to_binary(decimal_num)
        Convert a decimal number to its binary representation.

        Args:
```

```
            decimal_num (int): A non-negative integer to convert
to binary.

        Returns:            str: The binary representation of
the decimal number (without '0b' prefix).

        Raises:
            ValueError: If the input is negative.

        Example:
            >>> decimal_to_binary(10)
            '1010'
            >>> decimal_to_binary(255)
            '11111111'

    decimal_to_hexadecimal(decimal_num)
        Convert a decimal number to its hexadecimal
representation.

        Args:
            decimal_num (int): A non-negative integer to convert
to hexadecimal.

        Returns:            str: The hexadecimal
representation of the decimal number (without '0x' prefix).

        Raises:
            ValueError: If the input is negative.

        Example:
            >>> decimal_to_hexadecimal(255)
            'ff'
            >>> decimal_to_hexadecimal(4095)
            'fff'
```

```
FILE
c:\users\rohit\onedrive\documents\sru\ai_code\lab_9.1\conversion.p y
```

Command for html version:
Python -m pydoc -w conversion

Output(conversion.html):

```
index
conversion c:\users\rohit\onedrive\documents\sru\ai_code\lab_9.1\conversion.py

Conversion module for number system conversions.

This module provides functions to convert between different number systems
including decimal, binary, and hexadecimal representations.

Functions

    binary_to_decimal(binary_str)
        Convert a binary string to its decimal representation.

        Args:
          binary_str (str): A binary number as a string (e.g., '1010').

        Returns:
          int: The decimal equivalent of the binary number.

        Raises:
          ValueError: If the input contains non-binary characters.

        Example:
          >>> binary_to_decimal('1010')
          10
          >>> binary_to_decimal('11111111')
          255

    decimal_to_binary(decimal_num)
        Convert a decimal number to its binary representation.

        Args:
          decimal_num (int): A non-negative integer to convert to binary.

        Returns:
          str: The binary representation of the decimal number (without '0b' prefix).

        Raises:
          ValueError: If the input is negative.

        Example:
          >>> decimal_to_binary(10)
          '1010'
          >>> decimal_to_binary(255)
          '11111111'

    decimal_to_hexadecimal(decimal_num)
        Convert a decimal number to its hexadecimal representation.

        Args:
          decimal_num (int): A non-negative integer to convert to hexadecimal.

        Returns:
          str: The hexadecimal representation of the decimal number (without '0x' prefix).

        Raises:
          ValueError: If the input is negative.

        Example:
          >>> decimal_to_hexadecimal(255)
          'ff'
          >>> decimal_to_hexadecimal(4095)
          'fff'
```

Problem 5 – Course Management Module Task:
1. Create a module course.py with functions:

o add_course(course_id, name, credits) o
remove_course(course_id) o
get_course(course_id)

2. Add docstrings with Copilot.
3. Generate documentation in the terminal.
4. Export the documentation in HTML format and open it in a browser.

Prompt:

Create course.py module with add_course, remove_course, get_course functions using dictionary storage and proper docstrings.

Code:

```
"""

Course management module for storing and retrieving course information.

"""



# Dictionary to store courses with course_id as key

courses = {}



def add_course(course_id, course_name, credits):

"""

    Add a new course to the storage.


Args:
```

```
        course_id (str): Unique identifier for the course

        course_name (str): Name of the course

        credits (int): Number of credit hours


    Returns:

        bool: True if course added successfully, False if course_id already

exists

"""
```

```
    if course_id in courses:
        return False

    courses[course_id] = {

        "name": course_name,

        "credits": credits

    }

    return True
```

```python
def remove_course(course_id):

    """

    Remove a course from the storage.


    Args:

        course_id (str): Unique identifier for the course to remove


    Returns:

        bool: True if course removed successfully, False if course_id not found

    """

    if course_id in courses:

        del courses[course_id]

        return True

    return False



def get_course(course_id):

    """

    Retrieve course information by course_id.
```

```
    Args:
        course_id (str): Unique identifier for the course

    Returns:
        dict: Course information with 'name' and 'credits' keys, or None if not
found
    """
    return courses.get(course_id)

# Example usage:
if __name__ == "__main__":
    add_course("CS101", "Introduction to Computer Science", 4)
    add_course("MATH201", "Calculus I", 3)

    print(get_course("CS101"))  # Output: {'name': 'Introduction to Computer
Science', 'credits': 4}
    print(get_course("MATH201"))  # Output: {'name': 'Calculus I', 'credits': 3}
```

```
    remove_course("CS101")

    print(get_course("CS101"))  # Output: None
```

Output:

```
{'name': 'Introduction to Computer Science', 'credits': 4}
{'name': 'Calculus I', 'credits': 3}
None
```

Command for documentation in terminal:
python -m pydoc course

Output:

```
PS C:\Users\rohit\OneDrive\Documents\SRU\ai_code\lab_9.1> python
-m pydoc course
Help on module course:

NAME       course - Course management module for storing and
retrieving course information.

FUNCTIONS
    add_course(course_id, course_name, credits)
Add a new course to the storage.

        Args:            course_id (str): Unique identifier
for the course                   course_name (str): Name of
the course             credits (int): Number of credit hours
```

```
        Returns:            bool: True if course added
successfully, False if course_id already exists

    get_course(course_id)
        Retrieve course information by course_id.

        Args:
            course_id (str): Unique identifier for the course

        Returns:
            dict: Course information with 'name' and 'credits'
keys, or None if not found

    remove_course(course_id)
        Remove a course from the storage.

        Args:
            course_id (str): Unique identifier for the course to
remove

        Returns:
            bool: True if course removed successfully, False if
course_id not found

DATA
    courses = {}

FILE
c:\users\rohit\onedrive\documents\sru\ai_code\lab_9.1\course.py
```

**Command for HTML version: python -
m pydoc -w course**

```
PS C:\Users\rohit\OneDrive\Documents\SRU\ai_code\lab_9.1> python
-m pydoc -w course
wrote course.html
```

Output(course.html):

index
course c:\users\rohit\onedrive\documents\sru\ai_code\lab_9.1\course.py

Course management module for storing and retrieving course information.

Functions

add_course(course_id, course_name, credits)
Add a new course to the storage.

Args:
course_id (str): Unique identifier for the course
course_name (str): Name of the course
credits (int): Number of credit hours

Returns:
bool: True if course added successfully, False if course_id already exists

get_course(course_id)
Retrieve course information by course_id.

Args:
course_id (str): Unique identifier for the course

Returns:
dict: Course information with 'name' and 'credits' keys, or None if not found

remove_course(course_id)
Remove a course from the storage.

Args:
course_id (str): Unique identifier for the course to remove

Returns:
bool: True if course removed successfully, False if course_id not found

Data
courses = {}